Lecture 2

Span, Linear Independence, Basis, Dimension

Last lecture we defined a vector space as a set with addition and scalar multiplication satisfying the vector-space axioms. A subset of a vector space was a subspace exactly when it was closed under linear combinations. In this lecture we build from linear combinations the two most important concepts of the whole course. A *basis* is the shortest list of vectors from which every vector of the space is built in exactly one way; it is the abstract counterpart of a coordinate system. The number of vectors in a basis is the *dimension* — the description of the "size" of a vector space.

The use of these concepts is entirely concrete. Once we fix a basis, every abstract vector — a polynomial, a matrix, a quantum state — becomes a simple column of numbers, and every linear-algebra computation turns into arithmetic you already know. In quantum mechanics, choosing a basis is exactly what physicists call choosing a *representation.* The symbol $|\psi\rangle$ will here be simply a label for the vector ψ; we will meet Dirac notation in Lecture 7. Finite or discrete bases give ordinary coordinate columns. Infinite-dimensional spaces also occur in quantum mechanics. There a function describes the state instead of a column of numbers — physicists call it a wave function. We will meet such spaces in Lectures 13–14. By the end of this lecture you will be able to find bases, compute dimensions, pass from a vector to its coordinates and back, and count dimensions of sums and intersections.

Learning objectives

By the end of this lecture you will be able to:

- Compute the span of a list and recognise it as the smallest containing subspace.
- Test a list for linear independence using the linear-dependence lemma.
- Find a basis and dimension, and pass between a vector and its coordinate column.
- Apply the dimension formula $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$.

1 Span

Definition 2.1 (Span). The *span* of a list $v_1, \dots, v_m$ in V is the set of all their linear combinations,

$$\operatorname{span}(v_1, \dots, v_m) = \{a_1 v_1 + \cdots + a_m v_m : a_1, \dots, a_m \in \mathbb{F}\}.$$

By convention the span of the empty list is $\{0\}$.

Proposition 2.2 (The span is the smallest containing subspace). $\operatorname{span}(v_1, \dots, v_m)$ is a subspace of V, and it is contained in every subspace of V that contains $v_1, \dots, v_m$.

Proof. It contains 0 (take all scalars 0), and a sum or scalar multiple of linear combinations of the v_k is again such a combination, so the subspace test (Lecture 1) is met. If a subspace U contains every v_k, then it contains all their linear combinations by closure, i.e. $U \supseteq \operatorname{span}(v_1, \dots, v_m)$. ∎

Definition 2.3 (Spanning list, finite-dimensional). If $\operatorname{span}(v_1, \ldots, v_m) = V$ we say $v_1, \ldots, v_m$ *spans* V. A vector space is *finite-dimensional* if some finite list spans it, and *infinite-dimensional* otherwise.

Example 2.4 (Standard spanning lists). The list $e_1 = (1, 0, \ldots, 0), \ldots, e_n = (0, \ldots, 0, 1)$ spans $\mathbb{F}^n$, since $(x_1, \ldots, x_n) = x_1 e_1 + \cdots + x_n e_n$; so $\mathbb{F}^n$ is finite-dimensional. Likewise $1, t, \ldots, t^m$ spans $\mathcal{P}_m(\mathbb{F})$. By contrast $\mathcal{P}(\mathbb{F})$ is *infinite*-dimensional. An empty or all-zero finite list does not even span 1. Any other finite list has a maximal degree d among its nonzero members, so its combinations cannot produce t^{d+1}. ◁

Geometrically, the span of a single nonzero vector is a line through the origin, and the span of two non-parallel vectors in $\mathbb{R}^3$ is a plane through the origin (Figure 1): every linear combination stays inside it.

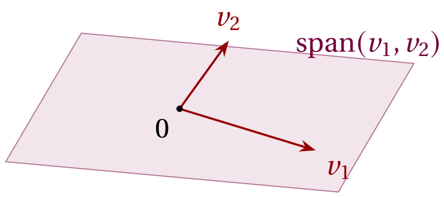

Figure 1: Two non-parallel vectors in $\mathbb{R}^3$ span a plane through the origin.

Example 2.5 (Is a vector in the span?). Is $(15, 16, 17)$ in the span $\operatorname{span}\big((1, 2, 3), (6, 5, 4)\big)$ in $\mathbb{R}^3$? We seek scalars a, b such that $a(1, 2, 3) + b(6, 5, 4) = (15, 16, 17)$. Comparing coordinates, we obtain the linear system

$$a + 6b = 15, \qquad 2a + 5b = 16, \qquad 3a + 4b = 17.$$

We solve it by Gaussian elimination. We write the vectors $(1, 2, 3)$ and $(6, 5, 4)$ into the columns of a matrix, the vector $(15, 16, 17)$ on the right-hand side, and reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 6 & 15 \\ 2 & 5 & 16 \\ 3 & 4 & 17 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{array}\right] \implies a = 3, \quad b = 2.$$

The last row is zero, so the system is consistent and has a unique solution. Hence $(15, 16, 17) = 3(1, 2, 3) + 2(6, 5, 4)$ and the answer is *yes*. The question whether a vector lies in a span is always the question whether a certain linear system is consistent. That is exactly why in Week 0 we said that Gaussian elimination would be needed everywhere. ◁

2 Linear independence

A spanning list may carry redundancy: some vector of it may already be a combination of the others. The most important lists are those with *no* redundancy.

Definition 2.6 (Linear independence). A list $v_1, \ldots, v_m$ is *linearly independent* if the only scalars $a_1, \ldots, a_m$ with

$$a_1 v_1 + \cdots + a_m v_m = 0$$

are $a_1 = \cdots = a_m = 0$. Otherwise it is *linearly dependent*: there exist scalars, not all zero, giving 0. (The empty list is independent.)

Remark 2.7. Independence says that each vector of the span has a *unique* expression as a linear combination: if $\sum a_k v_k = \sum b_k v_k$ then $\sum(a_k - b_k)v_k = 0$. This means that $a_k = b_k$. That is exactly why a basis will give every vector uniquely determined coordinates.

Example 2.8 (Independent and dependent lists). The list of standard unit vectors

$$e_1 = (1, 0, \dots, 0), \quad \dots, \quad e_n = (0, \dots, 0, 1)$$

is linearly independent in $\mathbb{F}^n$, as is $1, t, \dots, t^m$ in $\mathcal{P}_m(\mathbb{F})$. A length-one list is independent iff its vector is nonzero; a length-two list is independent iff neither vector is a scalar multiple of the other. An example of a dependent list in $\mathbb{F}^3$ is $(2, 3, 1)$, $(1, -1, 2)$, $(7, 3, 8)$, because

$$2(2, 3, 1) + 3(1, -1, 2) + (-1)(7, 3, 8) = (0, 0, 0).$$

From your calculus course you know that the three functions $1,\ \cos^2 t,\ \sin^2 t$ are linearly *dependent* in $\mathbb{R}^{\mathbb{R}}$, because $1 - \cos^2 t - \sin^2 t = 0$ for all $t \in \mathbb{R}$. ◁

Example 2.9 (Testing independence by a determinant). For which c are the vectors $(1, 2, 3)$, $(0, 1, 4)$, $(2, 3, c)$ independent in $\mathbb{R}^3$? Three vectors in $\mathbb{R}^3$ are independent exactly when the matrix with these rows is invertible, i.e. has nonzero determinant. Expanding along the first column,

$$\det\begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 2 & 3 & c \end{pmatrix} = 1\,(c - 12) - 0 + 2\,(8 - 3) = c - 12 + 10 = c - 2.$$

So the list is linearly independent precisely when $c \neq 2$, and dependent when $c = 2$. The determinant test (and, equivalently, row reduction) is the practical way to decide independence of vectors given by coordinates. ◁

The next lemma is the main tool of this lecture — almost all the results that follow rely on it. It says something simple: in a dependent list there is always a redundant vector. It can be "removed".

Lemma 2.10 (Linear dependence lemma). If $v_1, \dots, v_m$ is linearly dependent, then there is an index j with $v_j \in \operatorname{span}(v_1, \dots, v_{j-1})$, and removing v_j from the list leaves the span unchanged. (For $j = 1$ this reads $v_1 = 0$.)

Proof. Dependence gives scalars $a_1, \dots, a_m$, not all zero, with $\sum a_i v_i = 0$. Let j be the largest index with $a_j \neq 0$. If $j = 1$ then $a_1 v_1 = 0$ with $a_1 \neq 0$, so $v_1 = 0 \in \{0\} = \operatorname{span}(\,)$. If $j > 1$ then

$$v_j = -\tfrac{1}{a_j}\big(a_1 v_1 + \cdots + a_{j-1} v_{j-1}\big) \in \operatorname{span}(v_1, \dots, v_{j-1}).$$

For the span claim, any combination of $v_1, \dots, v_m$ becomes a combination of the remaining vectors once we substitute this expression for v_j; hence deleting v_j does not shrink the span (and clearly does not enlarge it). ■

Example 2.11 (Finding the redundant vector). Take $(1, 2, 3)$, $(6, 5, 4)$, $(15, 16, 17)$, $(8, 9, 7)$ in $\mathbb{R}^3$. We will soon see $\mathbb{R}^3$ is three-dimensional, so a list of four vectors must be dependent; the dependence lemma then says some vector is a combination of the earlier ones. Which is the first? Gaussian elimination answers this at once. We write all four vectors into the columns of a matrix, in the order of the list, and reduce the matrix:

$$\begin{bmatrix} 1 & 6 & 15 & 8 \\ 2 & 5 & 16 & 9 \\ 3 & 4 & 17 & 7 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The pivots are in the first, second and fourth columns, while the third column has no pivot. The numbers 3 and 2 written in it are precisely the coefficients we want: $(15, 16, 17) = 3(1, 2, 3) + 2(6, 5, 4)$ — the same answer we obtained in Example 2.5. Thus $j = 3$ is the first index at which a vector lies in the span of its predecessors. Deleting $(15, 16, 17)$ leaves the span unchanged.

This is always the case: a column with no pivot is a linear combination of the earlier columns. The number of that column gives the index j of Lemma 2.10, and its entries give the linear combination we are looking for. ◁

3 The fundamental counting theorem

Everything that follows — that dimension is precisely defined, that bases behave as coordinate systems — rests on one inequality.

> **Theorem 2.12 (Independent $\leq$ spanning).** In a vector space V, the length of any linearly independent list is at most the length of any spanning list.

Proof. Let $u_1, \dots, u_m$ be independent and $w_1, \dots, w_n$ span V; we prove $m \leq n$. We run a process of m steps, each inserting one u and deleting one w, keeping a spanning list of length n throughout.

Step 1. The list $w_1, \dots, w_n$ spans V, so adjoining u_1 in front gives the dependent list $u_1, w_1, \dots, w_n$ (here u_1 lies in the span of the w's). By Lemma 2.10 some entry is in the span of the earlier ones; it is not u_1, since independence makes $u_1 \neq 0$ and $\operatorname{span}(\,) = \{0\}$. So it is some w_k, which we delete. The result is a spanning list of length n containing u_1.

Step k $(2 \leq k \leq m)$. Suppose we hold a spanning list of length n containing $u_1, \dots, u_{k-1}$. Insert u_k just after u_{k-1}: the new list is dependent (every added vector lies in the span of a spanning list). By Lemma 2.10 some entry is in the span of the earlier ones. It cannot be any of $u_1, \dots, u_k$, for those are independent and so none lies in the span of the previous u's. Hence it is one of the remaining w's, which we delete, restoring a spanning list of length n that now contains $u_1, \dots, u_k$.

If $m > n$, then after Step n we hold a spanning list of length n consisting of $u_1, \dots, u_n$ *only* (all w's gone). Step $n+1$ inserts u_{n+1}, producing a dependent list of u's alone; Lemma 2.10 then puts some u in the span of earlier u's, contradicting independence. Therefore $m \leq n$. ∎

A useful consequence of this theorem is that every subspace U of a finite-dimensional space V is itself finite-dimensional. Choose one vector of U that does not lie in the span of those already chosen, and repeat this as long as it is possible. The chosen vectors stay independent throughout, since none of them lies in the span of its predecessors — this is exactly Lemma 2.10, only read in the opposite direction. Therefore Theorem 2.12 caps their number at $\dim V$, and the process must stop. And it can only stop when no vector of U lies outside the current span, that is, when the chosen list spans U.

4 Bases and dimension

> **Definition 2.13 (Basis).** A *basis* of V is a list that is linearly independent and spans V.

> **Theorem 2.14 (A basis gives uniquely determined coordinates).** A list $e_1, \dots, e_n$ is a

basis of V if and only if every $v \in V$ has a unique expression $v = a_1e_1 + \cdots + a_ne_n$ with $a_k \in \mathbb{F}$.

Proof. ($\Rightarrow$) Spanning gives the existence of an expression, and independence gives its uniqueness; we showed this in the remark following the definition of linear independence. ($\Leftarrow$) If such an expression exists for every v, then the list spans V. Applying uniqueness to the equality $0 = \sum a_ke_k$ (which also equals $\sum 0 \cdot e_k$) gives that every $a_k = 0$, and that is independence. So the list is a basis. ■

Example 2.15 (Standard bases). $e_1, \ldots, e_n$ is a basis of $\mathbb{F}^n$ (the *standard basis*); $1, t, \ldots, t^m$ is a basis of $\mathcal{P}_m(\mathbb{F})$; and the matrix units E_{ij} (a 1 in entry (i, j), zeros elsewhere) form a basis of $M_{m\times n}(\mathbb{F})$. ◁

A basis can always be produced in two complementary ways: by shortening a spanning list, or by extending an independent list.

Theorem 2.16 (Reduction to a basis). Every spanning list of V can be reduced to a basis of V by deleting some of its vectors.

Proof. While the list is linearly dependent, Lemma 2.10 supplies a vector lying in the span of the earlier ones; delete it, which by the same lemma leaves the span equal to V. Each deletion shortens a finite list, so the process halts, and it can halt only when the list is independent—an independent spanning list, i.e. a basis. ■

Theorem 2.17 (Extension to a basis). In a finite-dimensional space, every linearly independent list extends to a basis.

Proof. Let $u_1, \ldots, u_m$ be independent and $w_1, \ldots, w_n$ a basis. Apply Theorem 2.16 to the spanning list $u_1, \ldots, u_m, w_1, \ldots, w_n$. No u_i is ever deleted: the vectors preceding it are $u_1, \ldots, u_{i-1}$, which are independent, so u_i is not in their span and is not a candidate for removal. Only w's are discarded, leaving a basis that still contains $u_1, \ldots, u_m$. ■

Corollary 2.18 (Existence of a basis). Every finite-dimensional vector space has a basis—reduce any finite spanning list.

Example 2.19 (Reducing a spanning list). The list $(1, 0), (0, 1), (1, 1)$ spans $\mathbb{R}^2$ but is dependent, since $(1, 1) = (1, 0) + (0, 1)$. That last vector lies in the span of the earlier two, so the reduction of Theorem 2.16 deletes it, leaving the basis $(1, 0), (0, 1)$. ◁

Example 2.20 (Extending an independent list). To extend $(1, 1, 0, 0), (0, 1, 1, 0)$ to a basis of $\mathbb{R}^4$, append the standard basis $e_1, \ldots, e_4$ and drop redundant vectors as in Theorem 2.17. One outcome is

$$(1, 1, 0, 0),\ (0, 1, 1, 0),\ (1, 0, 0, 0),\ (0, 0, 0, 1),$$

which is linearly independent and spans $\mathbb{R}^4$, hence a basis. ◁

For dimension to make sense, all bases of a space must have the same length. Theorem 2.12 delivers exactly this.

Theorem 2.21 (Basis length is precisely defined). Any two bases of a finite-dimensional vector space have the same length.

Proof. Let $\mathcal{B}_1, \mathcal{B}_2$ be bases. Since $\mathcal{B}_1$ is independent and $\mathcal{B}_2$ spans, Theorem 2.12 gives $\operatorname{len} \mathcal{B}_1 \le \operatorname{len} \mathcal{B}_2$. Swapping their roles gives the reverse inequality, so the lengths are equal. ■

Definition 2.22 (Dimension). The *dimension* $\dim V$ of a finite-dimensional vector space is the length of any basis.

Example 2.23 (Dimensions). $\dim \mathbb{F}^n = n$; $\dim \mathcal{P}_m(\mathbb{F}) = m+1$; $\dim M_{m\times n}(\mathbb{F}) = mn$. And, as promised in Lecture 1, the field $\mathbb{C}$ has $\dim_{\mathbb{C}} \mathbb{C} = 1$ but $\dim_{\mathbb{R}} \mathbb{C} = 2$ (basis $1, i$): the dimension depends on the field of scalars, not just on the set. ◁

Two shortcuts make checking a basis far easier once the dimension is known; both follow from Theorem 2.12.

Proposition 2.24 (Right length is enough). Let $\dim V = n$. Then any linearly independent list of length n is a basis, and any spanning list of length n is a basis.

Proof. An independent list of length n extends to a basis by Theorem 2.17; but every basis has length n, so no vector is added and the list is already a basis. A spanning list of length n reduces to a basis by Theorem 2.16; again every basis has length n, so no vector is removed and the list is already a basis. ■

Proposition 2.25 (Subspaces have smaller dimension). If V is finite-dimensional and $U \subseteq V$ is a subspace, then $\dim U \le \dim V$, with equality if and only if $U = V$.

Proof. A basis of U is an independent list in V, so $\dim U \le \dim V$ by Theorem 2.12. If $\dim U = \dim V = n$, a basis of U is an independent list of length n in V, hence a basis of V by Proposition 2.24; thus $U = \operatorname{span}(\text{that list}) = V$. ■

Example 2.26 (A basis of a subspace by solving a system). Let

$$U = \{x \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0,\ x_1 - x_3 = 0\}.$$

Both defining equations are homogeneous, so U is the solution set of the homogeneous system $Ax = 0$ — the null space of the matrix A. We write the coefficients of the equations into the rows of a matrix and reduce the matrix A (there is no need to write the right-hand side, since it stays zero throughout):

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & -1 & 0 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 1 \end{bmatrix} \implies x_1 = x_3, \quad x_2 = -2x_3 - x_4.$$

The pivots are in the first and second columns, so x_1 and x_2 are the pivot variables and the free ones are $s = x_3$ and $t = x_4$. The free variables may take any values, and the pivot variables are then determined uniquely by the RREF. Let us write the general solution as a single column and split it along s and t:

$$x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} s \\ -2s-t \\ s \\ t \end{pmatrix} = s\begin{pmatrix} 1 \\ -2 \\ 1 \\ 0 \end{pmatrix} + t\begin{pmatrix} 0 \\ -1 \\ 0 \\ 1 \end{pmatrix}$$

$$\implies \quad U = \operatorname{span}\big((1,-2,1,0),\ (0,-1,0,1)\big).$$

The columns standing next to s and t are the basis vectors we want. Each of them is obtained by setting one free variable to 1 and all the others to 0: the vector $(1, -2, 1, 0)$ corresponds to $(s, t) = (1, 0)$, and $(0, -1, 0, 1)$ to $(s, t) = (0, 1)$. That is exactly why they are independent: in the positions of the free variables, that is, in the third and fourth coordinates, they carry $(1, 0)$ and $(0, 1)$, so neither is a multiple of the other. Hence these two vectors form a basis and $\dim U = 2$. This is exactly how we found a basis of a null space in Week 0. ◁

5 Coordinates relative to a basis

Theorem 2.14 lets us attach numbers to abstract vectors.

Definition 2.27 (Coordinate vector). Let $\mathcal{B} = (e_1, \ldots, e_n)$ be an *ordered* basis of V. The *coordinate vector* of $v = a_1e_1 + \cdots + a_ne_n$ is $[v]_{\mathcal{B}} = (a_1, \ldots, a_n) \in \mathbb{F}^n$.

Proposition 2.28 (The coordinate map). The map $v \mapsto [v]_{\mathcal{B}}$ is a bijection $V \to \mathbb{F}^n$ compatible with both operations: $[v + w]_{\mathcal{B}} = [v]_{\mathcal{B}} + [w]_{\mathcal{B}}$ and $[av]_{\mathcal{B}} = a\,[v]_{\mathcal{B}}$.

Proof. Bijectivity is Theorem 2.14 (existence and uniqueness of coordinates). Reading off coefficients is additive and homogeneous, because the basis expansion itself is. ■

This proposition joins the abstract theory to practical computation. Once a basis is chosen, *every* n-dimensional space over $\mathbb{F}$ looks like $\mathbb{F}^n$. We will name this phenomenon — *isomorphism* — in Lecture 4. The coordinates of v depend on the basis, as Figure 2 shows.

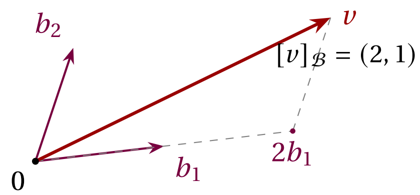

Figure 2: Coordinates relative to a (non-orthogonal) basis $\mathcal{B} = (b_1, b_2)$: reach v by going 2 steps along b_1 and 1 along b_2.

Example 2.29 (Coordinates of a polynomial). In $\mathcal{P}_2(\mathbb{R})$ take the polynomial $p(t) = 3+2t-t^2$. In the standard basis $\mathcal{B} = (1, t, t^2)$ we read $[p]_{\mathcal{B}} = (3, 2, -1)$. In the shifted basis $C = \big(1,\ t-1,\ (t-1)^2\big)$, substitute $s = t-1$ (so $t = s+1$):

$$p = 3 + 2(s+1) - (s+1)^2 = 3 + 2s + 2 - (s^2 + 2s + 1) = 4 - s^2,$$

so $[p]_C = (4, 0, -1)$. It is the same polynomial, only the coordinate columns differ. ◁

Example 2.30 (Coordinates by solving a system). Take the basis $\mathcal{B} = \big((1, 1), (1, -1)\big)$ of $\mathbb{R}^2$ and the vector $v = (4, 2)$. We seek scalars a, b such that $v = a(1, 1) + b(1, -1)$. We write the basis vectors into the columns of a matrix, the vector v on the right-hand side, and reduce the augmented matrix:

$$\left[\begin{array}{cc|c} 1 & 1 & 4 \\ 1 & -1 & 2 \end{array}\right] \xrightarrow{\text{RREF}} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 1 \end{array}\right] \implies a = 3, \quad b = 1.$$

So $[v]_{\mathcal{B}} = (3, 1)$, even though in the standard basis $[v] = (4, 2)$. Finding coordinates in a given basis is, once again, solving a linear system. ◁

Remark 2.31 (Bases as representations). Let us mention Dirac notation briefly, ahead of meeting it properly later: $|v\rangle$ denotes the same vector as v. In a finite or discrete basis an expansion reads $|\psi\rangle = \sum_k a_k |e_k\rangle$, and the coordinate vector $(a_1, \ldots, a_n)$ is the "wavefunction" of the state in that basis. In the continuous position or momentum representation the analogue is a function, and to describe it we need the Hilbert-space machinery of Lectures 13–14. Changing the representation changes the coordinates, but the pure state itself stays the same. Linear algebra provides the methods by which such changes are carried out "mechanically".

6 The dimension formula for sums

Recall from Lecture 1 that $U + W$ and $U \cap W$ are subspaces. Their dimensions are linked by a counting formula, similar to the inclusion–exclusion principle for sets.

Theorem 2.32 (Dimension of a sum). If U and W are subspaces of a finite-dimensional vector space, then

$$\dim(U + W) = \dim U + \dim W - \dim(U \cap W).$$

Proof. Choose a basis $v_1, \ldots, v_p$ of $U \cap W$, so $\dim(U \cap W) = p$. This basis is an independent list in U, so it extends to a basis $v_1, \ldots, v_p,\ x_1, \ldots, x_q$ of U (so $\dim U = p + q$), and to a basis $v_1, \ldots, v_p,\ y_1, \ldots, y_r$ of W (so $\dim W = p + r$). We claim

$$v_1, \ldots, v_p,\ x_1, \ldots, x_q,\ y_1, \ldots, y_r \tag{2.1}$$

is a basis of $U + W$, which finishes the proof since then $\dim(U + W) = p + q + r = (p + q) + (p + r) - p$.

The list (2.1) clearly spans $U + W$. Let us prove its independence. Suppose

$$\sum_i \alpha_i v_i + \sum_j \beta_j x_j + \sum_k \gamma_k y_k = 0.$$

Put $z = \sum_i \alpha_i v_i + \sum_j \beta_j x_j$; then $z \in U$ and also $z = -\sum_k \gamma_k y_k \in W$, so $z \in U \cap W$. Hence z is a combination of the v_i alone; comparing with $z = \sum_i \alpha_i v_i + \sum_j \beta_j x_j$ and using independence of the basis of U forces every $\beta_j = 0$. The original relation becomes $\sum_i \alpha_i v_i + \sum_k \gamma_k y_k = 0$, and independence of the basis of W forces all $\alpha_i = \gamma_k = 0$. Thus (2.1) is independent. ∎

Corollary 2.33 (Direct sums add dimensions). $\dim(U \oplus W) = \dim U + \dim W$. More generally, $U + W$ is direct if and only if $\dim(U + W) = \dim U + \dim W$. Both statements extend to finitely many summands by induction on m: $\dim(U_1 \oplus \cdots \oplus U_m) = \dim U_1 + \cdots + \dim U_m$.

Proof. By Theorem 2.32, $\dim(U + W) = \dim U + \dim W$ exactly when $\dim(U \cap W) = 0$, i.e. $U \cap W = \{0\}$, which is the two-subspace criterion for directness from Lecture 1. ∎

Example 2.34 (Two planes must meet). If U, W are two-dimensional subspaces (planes through 0) of $\mathbb{R}^3$, then $\dim(U \cap W) = \dim U + \dim W - \dim(U + W) \geq 2 + 2 - 3 = 1$. So two planes through the origin in $\mathbb{R}^3$ always intersect in at least a line—they can never meet only at 0. ◁

Example 2.35 (Computing $\dim(U + W)$ and $\dim(U \cap W)$). In $\mathbb{R}^4$ let

$$U = \operatorname{span}\big((1,1,0,0), (0,0,1,1)\big), \qquad W = \operatorname{span}\big((1,0,1,0), (0,1,0,1)\big),$$

both of dimension 2. We write all four spanning vectors into the rows of a matrix and reduce it:

$$\begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \xrightarrow{\text{RREF}} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Three nonzero rows remain, so only three of the four vectors are independent and $\dim(U + W) = 3$. The formula then gives

$$\dim(U \cap W) = \dim U + \dim W - \dim(U + W) = 2 + 2 - 3 = 1.$$

Indeed $(1, 1, 1, 1) = (1, 1, 0, 0) + (0, 0, 1, 1) \in U$ and also $(1, 1, 1, 1) = (1, 0, 1, 0) + (0, 1, 0, 1) \in W$, so $U \cap W = \operatorname{span}\big((1, 1, 1, 1)\big)$, confirming the count. ◁

Summary of main concepts

The span of a list is the smallest subspace containing it. A list is *independent* when 0 has only the trivial representation, and a *basis* is an independent spanning list — or, what is the same, a list giving every vector uniquely determined coordinates. Every linearly independent list is no longer than any spanning list (Theorem 2.12); hence all bases have the same length, and that length is called the *dimension*. In an n-dimensional space, an independent or spanning list of length n is automatically a basis. Once an ordered basis is fixed, each vector turns into a coordinate column in $\mathbb{F}^n$; in a finite (discrete) model this is what a physicist calls choosing a representation. Continuous position and momentum representations instead produce wavefunctions and require the later Hilbert-space framework. Finally, $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$, with equality to $\dim U + \dim W$ exactly for direct sums.

- **Span**—all linear combinations of a list; the smallest subspace containing it.
- **Linear independence**—only the trivial combination yields 0.
- **Basis**—an independent spanning list; gives every vector uniquely determined coordinates.
- **Dimension**—the common length of all bases (independent ≤ spanning).
- **Coordinate vector**—the column $[v]_{\mathcal{B}} \in \mathbb{F}^n$ once a finite ordered basis is fixed; continuous representations are function-valued analogues.
- **Dimension formula**—$\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$; additive exactly for direct sums.

The problems for this lecture are in the sheet `exercises-2.pdf`.